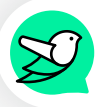


BostonDynamics



Senior Staff Manufacturing Engineer

[Apply](#) Waltham Office (POST) Full time Posted 4 Days Ago R2966

Sr. Staff Manufacturing Engineer

About Boston Dynamics:

Boston Dynamics is a global leader in robotics innovation, renowned for developing highly advanced, dynamic robots that excel in mobility, dexterity, and intelligence. Our cutting-edge robots, including the iconic Atlas, Spot, and Stretch, are pushing the boundaries of automation and redefining how machines interact with the world. At Boston Dynamics, we blend advanced mechanical design, state-of-the-art control systems, and AI-driven perception to create robots capable of performing in unstructured, real-world environments.

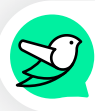
We are looking for an expert Sr. Staff Manufacturing Engineer to support the Atlas program, helping to transform groundbreaking research prototypes into reliable, scalable systems for manufacturing and field deployment.

About the Role:

As a Sr. Staff Manufacturing Engineer, you will play a critical role in the optimization and scale-up of low-volume production processes, assembly workflows, and custom tooling to support the assembly, test, and integration of our next-generation humanoid robot. You will work closely with Hardware Engineering, Prototype Operations, Systems Engineering, Behavior and Software teams to ensure that cutting-edge designs can be efficiently and reliably built in a prototype and low-volume production environment.

This role requires expert-level technical expertise in cradle-to-grave process implementation, large production system/value-stream design, and an innovative spirit to bridge the gap between product design and

manufacturing execution. You will oversee automation equipment, fixture, tool, and jig design and implementation to ensure process repeatability and product quality for Atlas builds. This role will also interface with our parent company, Hyundai Motor Group (HMG), leveraging their deep manufacturing expertise in the automotive sector.

**Responsibilities:**

- Develop, optimize, and implement rate production processes and assembly workflows for Atlas, from prototype through low-volume production with a focus on scaling for future production needs.
- Design and implement production system tools including material flow diagrams, value stream maps, floor layouts, and automation integrations.
- Collaborate with Hardware Engineering to provide DFM/DFA (Design for Manufacturing & Assembly) input throughout product development, ensuring manufacturability and serviceability.
- Design, fabricate, and validate custom fixtures, tools, and jigs to support mechanical assembly, precision alignment, and testing of electric motors and actuators, ensuring ergonomic and repeatable processes.
- Define assembly sequences, work instructions, and process documentation, including standard operating procedures (SOPs), visual aids, and training materials for assembly technicians.
- Support prototype builds in a hands-on capacity, troubleshooting assembly issues, identifying root causes, and driving corrective actions to improve processes and product designs.
- Implement process controls to ensure build quality, dimensional accuracy, and alignment with product performance requirements.
- Develop and implement inspection and test processes, including first article inspections (FAI), in-process quality checks, and final acceptance criteria.
- Collaborate with Supply Chain to ensure external fabrication and supplier processes align with internal manufacturing requirements and quality standards.
- Lead continuous improvement initiatives, applying Lean Manufacturing and Six Sigma methodologies to improve process efficiency, reduce cycle times, and increase yield.
- Drive tooling and equipment validation, ensuring custom fixtures and tools are fully qualified for use in production, including performing capability studies where necessary.

Qualifications:

- Bachelor's degree in Manufacturing, Mechanical, Industrial, Electrical Engineering, or a related field; Master's degree or PhD a plus.

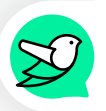


- At least 8-15 years of relevant experience in Process engineering, process development, or advanced manufacturing, preferably in robotics, aerospace, or complex electromechanical systems.
- Expert-level proficiency in CAD software (SolidWorks, CATIA, or similar) for tool/fixture design and review of engineering drawings.
- Proven track record of DFM/DFA collaboration with design teams to drive manufacturable and serviceable product designs.
- Hands-on experience developing and validating custom tooling, jigs, and fixtures to support precision mechanical assembly.
- Experience developing and validating custom automation equipment to support precision mechanical assembly and improve process efficiency.
- Experience writing clear, concise, and detailed work instructions, SOPs, and training materials for assembly technicians.
- Familiarity with precision assembly techniques, including alignment, shimming, fastening, sealing, and torque control for complex assemblies.
- Experience working in prototype, low-volume, or high-mix production environments, with an ability to adapt processes for evolving product designs.
- Strong understanding of GD&T (Geometric Dimensioning & Tolerancing) and its application in both design and inspection processes.
- Demonstrated experience applying Lean Manufacturing, Six Sigma, and root cause analysis methodologies to drive process improvement.
- Excellent problem-solving skills with the ability to troubleshoot complex mechanical and electromechanical assemblies.
- Comfortable working in cross-functional teams, balancing technical, schedule, and cost considerations in a fast-paced environment.
- Strong communication skills, with the ability to present technical concepts and influence stakeholders across engineering, operations, and leadership teams.
- Experience supporting on-site prototype builds and providing direct manufacturing floor support.
- Hands-on experience with additive manufacturing, rapid prototyping, machining (e.g. Mill, Lathe) and fabrication (e.g. Welding, Grinding, Bending) techniques for fixture development.

Preferred Qualifications

- Experience developing and documenting CNC machining processes for novel materials and manufacturing methods including specialized gear manufacturing.

- Experience supporting humanoid or robotic development programs.
- Familiarity with precision motion systems, actuators, and advanced sensors.
- Experience with Product Lifecycle Management (PLM) systems for engineering change control and manufacturing documentation management.
- Experience with Manufacturing Execution Systems (MES) for production floor management, process control and visual management.
- Background working with external design firms, contract manufacturers, or suppliers for tool and fixture development.
- Experience working in a research-to-product transition environment, where designs and processes evolve rapidly.
- TWI Job Instruction and/or Job Methods certified



We are interested in every qualified candidate who is eligible to work in the United States. However, we are not able to sponsor visas for this position.

The salary or hourly pay range for this position will be clearly stated in the job posting as required by Massachusetts law. The base pay range for this position is between \$135,000.00– \$170,000. Base pay will depend on multiple individualized factors including, but not limited to internal equity, job related knowledge, skills and experience. This range represents a good faith estimate of compensation at the time of posting.

Boston Dynamics will never ask you to divulge your personal financial or account information as part of its recruiting process.

About Us

We are growing rapidly, building a commercial company that delivers cutting edge technology and solutions to our customers from industrial applications to logistics and warehouse solutions.

[Read More](#) ✓

